AI Anxiety Map: Geographic and Network Shifts in Global Labor-Market News, 2021–2024

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***Abstract*—Generative artificial intelligence (AI) has intensified debate about job displacement and new economic opportunities, but global evidence about how this debate is framed remains limited. This study analyzes labor-market news in the GDELT Global Knowledge Graph from 2021 through 2024. It compares two equal periods around January 1, 2023 using anxiety and opportunity scores constructed from Global Content Analysis Measures, geographic summaries, and location-emotion similarity networks. Aggregated location-theme article occurrences increased by 13.9% after the cutoff, while mean anxiety and opportunity scores each decreased by about 6.9%. This global reduction was not geographically uniform: anxiety increased in 122 of 197 country summaries and in 46 of 50 U.S. states. Country changes in anxiety and opportunity were positively correlated ($r = 0.57$), indicating that they often represent common variation in emotional intensity rather than opposite ends of one scale. Among 18 high-volume locations shared by both network periods, mean pairwise emotional similarity decreased by 1.5%, with 80 pairs converging and 73 diverging. The findings demonstrate the value of combining temporal, geographic, and network views, while describing news discourse rather than individual attitudes or causal labor-market effects.**

***Index Terms*—generative AI, labor market, GDELT, emotion analysis, geographic analysis, network analysis**

I. Introduction

Generative AI can automate parts of knowledge work while also complementing workers and creating new tasks. Earlier research describes the same tension for automation more broadly: technology replaces some tasks, increases the value of complementary skills, and distributes adjustment costs unevenly [1], [2]. Recent experiments report productivity gains from generative AI in professional writing and customer support [3], [4]. Such gains can coexist with concern about displacement, wage pressure, and skill obsolescence.

News media provide a large, time-stamped record of how these competing possibilities are framed. They are not a survey of workers, but they make it possible to compare discourse across countries with a common procedure. Prior longitudinal analysis has shown that news coverage of AI contains persistent optimism together with increasing concern about ethics, control, and employment [5]. The present project extends that approach to a multinational news index and explicitly models geography and cross-location similarity.

The study asks four research questions:

1) **RQ1:** How did the volume and emotional framing of global labor-market news change between 2021–2022 and 2023–2024?
2) **RQ2:** To what extent did those changes vary across countries and U.S. states?
3) **RQ3:** Did anxiety- and opportunity-related framing behave as opposing dimensions, or did they co-vary across locations?
4) **RQ4:** How did the emotional-similarity network among high-volume locations change after the cutoff, and which location pairs converged or diverged most?

The contribution is a reproducible framework that combines dictionary-based emotion measures, weekly and period comparisons, geographic summaries, and weighted networks. January 1, 2023 is an operational boundary for the period of broad generative-AI diffusion; it is not treated as a causal intervention.

II. Related Work

Research on automation emphasizes both substitution and complementarity. Autor argues that automation changes task composition rather than eliminating occupations uniformly [1], while Acemoglu and Restrepo document negative employment and wage effects in U.S. labor markets more exposed to industrial robots [2]. Evidence on generative AI likewise combines productivity gains with heterogeneous effects across tasks and worker experience [3], [4]. These findings motivate separate measures of anxious and opportunity-oriented framing.

Computational media studies use transparent dictionaries to measure affective language at scale. LIWC maps words to psychologically meaningful categories [8], and WordNet-Affect attaches emotion labels such as fear, hope, and optimism to lexical concepts [9]. The GDELT Global Knowledge Graph (GKG) indexes themes, locations, tone, and Global Content Analysis Measures (GCAM) from international news [6], [7]. Two-mode networks are appropriate when one node class represents locations and another represents their emotion attributes [10]. This study joins these strands by examining both geographic scores and changes in the projected location network.

III. Methodology

A. Collection and Aggregation

The pipeline reads the main and translated GDELT 2.1 GKG streams. A record is retained when its structured `V2Themes` field matches at least one configured labor-market term, including employment, unemployment, jobs, workers, hiring, layoffs, wages, income, careers, unions, gig work, and freelance work. For each retained record, the pipeline extracts the date, matched themes, all structured locations, GDELT tone, and selected GCAM dimensions. Records without a parsed location receive an unknown-location label.

One GKG record can mention several locations and match several labor themes. It therefore produces one interaction for every location-theme combination. The pipeline aggregates these interactions into seven-day windows by location, country and administrative codes, and theme. Consequently, summed `article_count` values are grouped article occurrences and can count one source record in multiple groups; they are not globally unique articles.

B. Emotional Scores and Period Comparison

For aggregate row i, the anxiety score is the mean of the available LIWC anxiety, WordNet anxiety, WordNet fear, and GCAM c6.6 uncertainty dimensions:

$$A_i = \frac{1}{|\mathcal{N}_i|} \sum_{d \in \mathcal{N}_i} x_{i,d}. \tag{1}$$

The opportunity score averages the available WordNet hope and optimism dimensions:

$$O_i = \frac{1}{|\mathcal{Q}_i|} \sum_{d \in \mathcal{Q}_i} x_{i,d}. \tag{2}$$

The anxiety-opportunity gap is $F_i = A_i - O_i$. A high score describes greater lexical prevalence in an aggregate news group, not clinical anxiety or a respondent's attitude. Scores are unweighted row means; volume is summarized separately. Because F_i is available only where both component scores are observed, its period mean need not equal the difference between the two marginal period means.

Rows whose window starts before January 1, 2023 are assigned to the before-AI period, and all later rows to the after-AI period. Global differences are assessed with Welch tests, while weekly series are used to inspect whether a period average conceals trends or peaks. Country and U.S. state summaries are computed only for units observed with both scores in both periods. Pearson correlation between country-level changes in A and O evaluates RQ3.

C. Location-Emotion Networks

For each period, a weighted bipartite graph links up to 20 countries with the largest interaction volumes to nine GCAM emotion dimensions. An edge weight is the mean score for that location-emotion combination. The projected similarity between locations u and v is

$$S(u,v) = \sum_{e \in E(u) \cap E(v)} \min\{w(u,e), w(v,e)\}, \tag{3}$$

where $w(u,e)$ is the bipartite edge weight. This overlap measure is larger when two locations share emotions at similar intensities. Percentage changes are calculated for location pairs present in both periods. Because the projection is complete and S is not normalized, this study interprets changes as network reconfiguration rather than as a direct polarization index.

TABLE I
Global before-after summary

Period	Occurrences	A	O	F
Before AI	718.9M	6.341	2.785	4.617
After AI	818.8M	5.902	2.592	4.225
Change (%)	+13.9	-6.9	-6.9	-8.5

IV. Experiments and Results

A. Dataset and Configuration

The production run covers January 1, 2021 through December 31, 2024, split into equal two-year periods. Each period contains 265 distinct GDELT location-country codes. The analysis is implemented in Python 3.11 with pandas, SciPy, NetworkX, Matplotlib, Plotly, and PyDeck. Scripts and notebooks are paired with Jupytext; fixed dates, a pinned Conda environment, checkpointed collection, and date-specific result directories support reproducibility. Large-scale collection and notebook execution were run on the Ben-Gurion University Slurm cluster.

B. RQ1: Global and Temporal Change

The weekly series in Fig. 1 does not resemble a single abrupt break at the cutoff. The largest anxiety peak was 12.017 in July 2022, before the cutoff; the largest after-period peak was 7.871 in May 2023. Anxiety then remained mostly below earlier levels during 2024. Opportunity followed a smoother decline with occasional joint spikes.

Table I and Fig. 2 show a combination of greater coverage and lower average emotional intensity. Grouped article occurrences rose by 99.9 million (13.9%), while anxiety and opportunity each fell by about 6.9%. The mean row-level anxiety-opportunity gap fell by 8.5%. Welch tests reject equal period means for anxiety ($t = -88.44$), opportunity ($t = -124.40$), and the gap ($t = -64.04$), all $p < .001$. Given the very large, clustered aggregate sample, effect magnitudes and temporal patterns are more informative than these small p-values. Thus, RQ1 is answered by increased discourse volume but lower average anxiety and opportunity intensity, embedded in substantial event-like temporal variation.

C. RQ2–RQ3: Geographic Heterogeneity and Co-Movement

The global decline masks pronounced geographic heterogeneity (Table II). Anxiety increased in 122 of 197 country summaries and decreased in 75; opportunity increased in 78 and decreased in 119. The U.S. state analysis points in a still different direction: the mean state anxiety score rose from 6.110 to 7.547, opportunity fell from 3.198 to 2.725, and

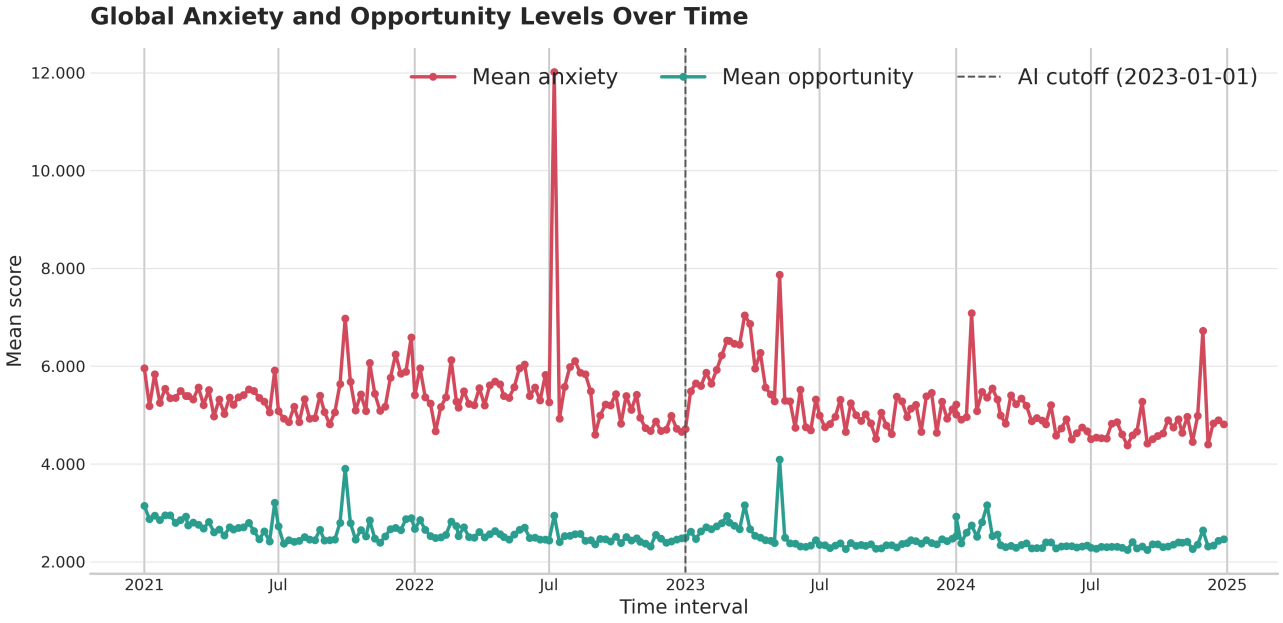


Fig. 1. Weekly global anxiety and opportunity scores. The dashed line marks the operational cutoff; the pre-cutoff peak cautions against a causal reading of the two-period comparison.

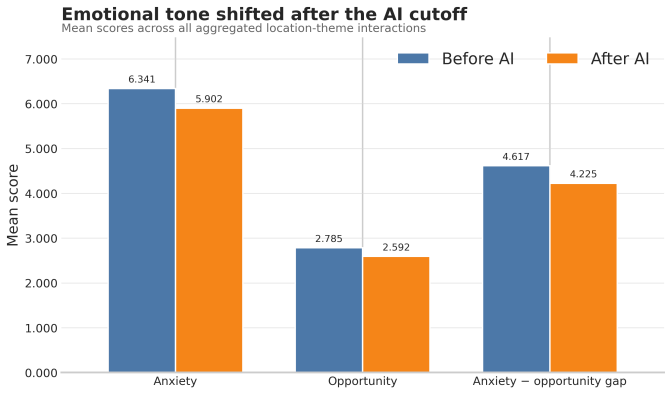


Fig. 2. Mean anxiety, opportunity, and row-level anxiety-opportunity gap before and after the cutoff.

TABLE II
DIRECTION OF GEOGRAPHIC SCORE CHANGES

Unit	N	$A+$	$A-$	$O+$	$O-$	Same	Opp.
Countries	197	122	75	78	119	137	60
U.S. states	50	46	4	17	33	21	29

anxiety increased in 46 of 50 states. These sign counts are unweighted by news volume, whereas global scores average all available aggregate rows. Their contrast therefore reveals both genuine heterogeneity and sensitivity to aggregation and coverage.

Anxiety and opportunity should not be treated as exact opposites. Across countries, their changes were positively correlated ($r = 0.572$): both rose in 70 countries and both fell in 67, so 137 of 197 countries moved in the same direction. Only 60 moved in opposite directions. RQ3 therefore identifies a shared emotional-intensity component as well as a balance between negative and positive framing. The row-level gap F is useful, but the two component scores remain necessary for interpretation.

D. RQ4: Location-Similarity Reconfiguration

The period networks each contain 20 locations and all 190 possible pairwise edges; 18 locations occur in both periods, producing 153 directly comparable pairs. Among these shared pairs, mean similarity declined from 49.326 to 48.581 (1.5%). The aggregate change was small, but individual relationships were mixed: 80 pairs became more similar and 73 became less similar.

The largest convergence was Russia–Ukraine (+97.7%), followed by Israel–Russia (+74.9%) and Israel–Ukraine (+72.7%). The largest divergence was China–Germany (-30.9%), followed by China–Vietnam (-26.8%) and Germany–Vietnam (-26.8%). Figure 3 visualizes the complete weighted networks. Node positions aid layout only; conclusions are based on edge weights and the matched-pair comparison. RQ4 is therefore best answered as selective reorganization, not uniform convergence or polarization. The largest changes also involve locations affected by major non-AI events, reinforcing the need for cautious attribution.

V. DISCUSSION

The four views prevent a misleading one-number conclusion. Globally, more labor-market news was indexed after the cutoff, but both anxiety and opportunity language weakened. Across geographic units, however, anxiety often

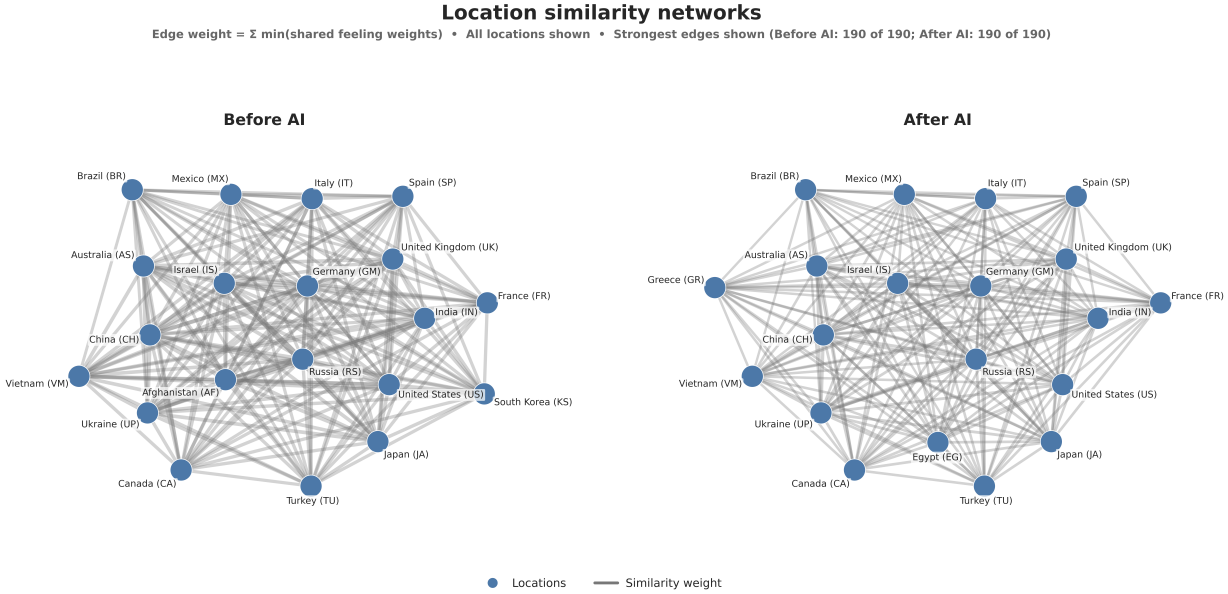


Fig. 3. Complete emotional-similarity networks for the 20 highest-volume locations in each period. Edge width represents the overlap score; two locations enter and two leave the high-volume set.

increased, particularly in U.S. states. The apparent disagreement is consistent with heterogeneous local trajectories and with differences between row-weighted global summaries and unweighted counts of location-level directions. The positive country-level correlation further shows that anxious and hopeful framing can intensify together when labor issues receive greater emotional attention.

The network results add relational information that country rankings cannot provide. A small decline in average similarity coexisted with very large changes for a few pairs. The strongest convergences cluster around Russia, Ukraine, and Israel, suggesting that geopolitical coverage is an important alternative explanation. The analysis therefore characterizes labor-market news during the generative-AI era, rather than isolating an effect of AI.

A. Limitations

GDELT coverage differs across countries, languages, and media systems, and an extracted location need not be the community affected by a story. Theme matching identifies labor-market discourse but does not require that every record explicitly discuss generative AI. Translation and dictionary methods can miss context, negation, and sarcasm, while GCAM dimensions may have different scales and missingness. Aggregated rows and repeated location-theme occurrences are not independent articles. The Welch tests do not model temporal or spatial dependence, and the before-after design cannot separate AI from wars, elections, macroeconomic changes, or shifts in source coverage. Finally, complete, unnormalized overlap networks do not directly measure polarization.

B. Future Work

Future analysis should require joint AI and labor themes, deduplicate at the article level, normalize emotion dimensions, and add coverage thresholds and uncertainty intervals for locations. Interrupted time-series or event-study models could test whether January 2023 marks a discontinuity. Normalized cosine similarity, community detection, and within- versus between-cluster dispersion would support a formal polarization question. Linking the news measures to surveys and labor indicators would help validate whether media framing corresponds to public attitudes or economic outcomes.

VI. Conclusion

This study presents a reproducible geographic and network framework for analyzing global labor-market news around the rise of generative AI. From 2021–2022 to 2023–2024, grouped article occurrences increased while global anxiety and opportunity averages declined. Most country summaries and nearly all U.S. states nevertheless recorded higher anxiety, and anxiety and opportunity commonly moved together. Location similarity changed selectively, with a modest average decline but substantial pair-specific convergence and divergence. These results show why temporal, geographic, and relational views must be interpreted together and why news discourse should not be confused with causal employment effects or individual perceptions.

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